1. <u>ENABLE OMR (E)</u>:

    Ctrl R ESC E Ctrl R E

2. <u>DISABLE OMR (D)</u>:

    Ctrl R ESC D Ctrl R E

3. <u>EXAMINE MEMORY LOCATIONS (EXAMINExxxxxx)</u>:

    Ctrl R ESC EXAMINExxxxxx Ctrl R E

    Where xx = Number of bytes to examine
    xxxx = Start address in hex

4. <u>EJECT TICKET OUT THE REAR (ER)</u>:

    Ctrl R ESC E R Ctrl R E

    NOTE: Use when EXIT CONTROL is enabled.

5. <u>EJECT TICKET OUT THE FRONT (EF)</u>:

    Ctrl R ESC E F Ctrl R E

    NOTE: Use when EXIT CONTROL is enabled.

6. <u>CHECK READER'S CONFIGURATION STATUS (GETCONFIG)</u>:

    Ctrl R ESC GETCONFIG Ctrl R E

    NOTE: Example of returned status is

    22 00 80 EVEINL80 1.5

Where 2	=	Baud rate constant prescaler (2 is 9.8304MHZ)
2	=	Selected baud rate: 0 = 38400 baud 1 = 19200 2 = 9600 3 = 4800 4 = 2400 5 = 1200 6 = 600 7 = 300
00	=	EEPROM Config. Flags status
80	=	Bottom timing
EVEINL80	=	Even parity, Inline timing, 80% threshold
1.5	=	1.5% decay

7. CHECK READER'S THRESHOLD AND DECAY SETTING (GETTBLS):

Ctrl R ESC GETTBLS Ctrl R E

NOTE: Example of returned status is

80 1.5

Where 80 is the threshold and 1.5 is the decay.

8. PROGRAM EE BYTE (PROGRAMxxxxxx) **(FOR FACTORY USE ONLY)**:

SYNTAX: Ctrl R ESC PROGRAMxxxxxx Ctrl R E
Ctrl R ESC CHECKSUM Ctrl R E

Where xx = Hex value to be programmed into EEPROM
xxxx = EEPROM address in hex

9. RESET THE READER (RESET):

Ctrl R ESC RESET Ctrl R E

10. ENTER SHOESHINE MODE (SHOE):

Ctrl R ESC SHOE Ctrl R E

NOTE: To exit the SHOESHINE mode, do a Shoeshine Stop command, Reset the reader, or power down.

11. EXIT SHOESHINE MODE (STOP):

Ctrl R ESC STOP Ctrl R E

12. SELECT BAUD RATE (SETBAUDxx):

SYNTAX:

Ctrl R	ESC	SETBAUDxx	Ctrl R	E	: Select baud rate
Ctrl R	ESC	CHECKSUM	Ctrl R	E	: Generate checksum
Ctrl R	ESC	RESET	Ctrl R	E	: Saves to EE

Where xx =

00 (hex)	= 38400 baud
01	= 19200 baud
02	= 9600 baud
03	= 4800 baud
04	= 2400 baud
05	= 1200 baud
06	= 600 baud
07	= 300 baud

13. SELECT DECAY RATE (SETDECAYx):

SYNTAX:	Ctrl R	ESC	SETDECAYx	Ctrl R	E	: Select decay rate
	Ctrl R	ESC	CHECKSUM	Ctrl R	E	: Generate checksum

Where x = 0 = 0.025%
1 = 0.05%
2 = 1.0%
3 = 1.5%
4 = 2%
5 = 4%

14. SELECT THRESHOLD (SETTHRESHx):

SYNTAX:	Ctrl R	ESC	SETTHRESHx	Ctrl R	E	: Select threshold rate
	Ctrl R	ESC	CHECKSUM	Ctrl R	E	: Generate checksum

Where x = 0 = 64%
1 = 66%
2 = 68%
3 = 70%
4 = 72%
5 = 74%
6 = 76%
7 = 78%
8 = 80%
9 = 82%

15. SELECT TIMING TYPE (SETTMTYPEx):

SYNTAX:	Ctrl R	ESC	SETTMTYPEx	Ctrl R	E	: Select timing type
	Ctrl R	ESC	CHECKSUM	Ctrl R	E	: Generate checksum

Where x = 0 = Inline
1 = Offset

16. SELECT TIMING CHANNEL (SETTMCHx):

SYNTAX:	Ctrl R	ESC	SETTMCHx	Ctrl R	E	: Select timing channel
	Ctrl R	ESC	CHECKSUM	Ctrl R	E	: Generate checksum
	Ctrl R	ESC	RESET	Ctrl R	E	: Saves to EEPROM

Where x = 0 = Bottom Timing
1 = Top Timing
2 = Both

17. CONFIGURE EEPROM FLAGS BYTE (SETFLAGSxx):

SYNTAX:	Ctrl R	ESC	SETFLAGSxx	Ctrl R	E	: Set EEPROM flags byte
	Ctrl R	ESC	CHECKSUM	Ctrl R	E	: Generate checksum
	Ctrl R	ESC	RESET	Ctrl R	E	: Saves to EEPROM

Where xx =
01 (hex) =Exit Control
02 = Feed Control
04 = Flow Control
08 = Flow Type
10 = Hollerith Mode
20 = Int XON
40 = True Data
80 = Reciprocate

EEPROM FLAGS BYTE DEFINITION

BIT 0	EXIT CONTROL	1 = Enabled (Need to use Eject Rear or Eject Front command to eject media. See example below) 0 = Disabled
BIT 1	FEED CONTROL	1 = Enabled manual-feed-control mode, need to give an Enable command before each feed, see example below) 0 = Disabled
BIT 2	FLOW CONTROL	1 = Enabled (need to also select RTS/CTS or Xon/Xoff, see FLOW TYPE below) 0 = Disabled
BIT 3	FLOW TYPE	1 = RTS/CTS Enabled 0 = Xon/Xoff Enabled
BIT 4	HOLLERITH MODE	1 = Enabled 0 = Disabled
BIT 5	INT XON	1 = Enabled 0 = Disabled
BIT 6	TRUE DATA	1 = Enabled 0 = Disabled
BIT 7	RECIPROCATE	1 = Enabled 0 = Disabled

<u>EXAMPLES</u>:

17.1 <u>DOCUMENT EXIT CONTROL ENABLED (SET BIT0 = 1)</u>:

Use SETFLAGSxx SYNTAX above and set BIT0 = 1

NOTE: Need to use Eject Rear command (Ctrl R ESC E R Ctrl R E)
or Eject Front command (Ctrl R ESC E F Ctrl R E) to eject media.

17.2 <u>DOCUMENT EXIT CONTROL DISABLED (SET BIT0 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT0 = 0

17.3 <u>DOCUMENT FEED CONTROL ENABLED (SET BIT1 = 1)</u>
<u>(Change to Manual Feed Control Mode)</u>:

Use SETFLAGSxx SYNTAX above and set BIT1 = 1

Need to give an Enable command (Ctrl R ESC E Ctrl R E) before each feed.

17.4 <u>DOCUMENT FEED CONTROL DISABLED (SET BIT1 = 0)</u>
<u>(Exit Manual Feed Control Mode)</u>:

Use SETFLAGSxx SYNTAX above and set BIT1 = 0

17.5 <u>ENABLE XON / XOFF DOCUMENT FLOW CONTROL</u>
<u>(SET BIT2 = 1 AND BIT3 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT2 = 1, BIT3 = 0

Use ‘CTRL Q’ for XON (ENABLE), and ‘CTRL S’ for XOFF (DISABLE)

17.6 <u>DISABLE XON / XOFF DOCUMENT FLOW CONTROL (SET BIT 2 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT 2 = 0

17.7 <u>ENABLE RTS / CTS FLOW TYPE (SET BIT2 = 1 AND BIT3 = 1)</u>:

Use SETFLAGSxx SYNTAX above and set BIT2 = 1, BIT3 = 1

This will set the Flow Type to RTS/CTS.

17.8 <u>DISABLE RTS / CTS FLOW TYPE (SET BIT2 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT2 = 0

17.9 <u>ENABLE HOLLERITH MODE (SET BIT4 = 1)</u>:

Use SETFLAGSxx SYNTAX above and set BIT4 = 1

17.10 <u>DISABLE HOLLERITH MODE (SET BIT4 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT4 = 0

NOTE: This will set the OMR back to BINARY MODE.

17.11 <u>ENABLE INITIAL XON (SET BIT5 = 1)</u>:

Use SETFLAGSxx SYNTAX above and set BIT5 = 1

NOTE: Need to also set BIT2 = 1 and BIT3 = 0 to enable XON / XOFF document flow control mode.
Use 'Ctrl S' for XOFF (DISABLE).

17.12 <u>DISABLE INITIAL XON (SET BIT5 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT5 = 0

17.13 <u>ENABLE TRUE DATA MODE (SET BIT6 = 1)</u>:

Use SETFLAGSxx SYNTAX above and set BIT6 = 1

17.14 <u>DISABLE TRUE DATA MODE (SET BIT6 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT6 = 0

This will set the OMR back to BINARY MODE

17.15 <u>ENABLE RECIPROCATE MODE (SET BIT7 = 1)</u>:

Use SETFLAGSxx SYNTAX above and set BIT7 = 1

17.16 <u>DISABLE RECIPROCATE MODE (SET BIT7 = 0)</u>:

Use SETFLAGSxx SYNTAX above and set BIT7 = 0

18 RESTORE FACTORY DEFAULTS:

Ctrl R ESC SETFACTORY Ctrl R E

19 SELECT PARITY:

Ctrl R ESC SETPARITYx Ctrl R E
Ctrl R ESC CHECKSUM Ctrl R E

Where x = 0 = Even
1 = Odd
2 = None

20. SET TEMP TIMMING CHANNEL (Sta):

Ctrl R ESC STA Ctrl R E

If the OMR's timing channel is set different than BOTTOM TIMING, this command will force the OMR's timing channel to bottom. This command is not saved in the EEPROM and will go away if the reader is reset.

21. CHECK READER'S STATUS:

Ctrl R ESC S Ctrl R E

NOTE: The reader will return a status byte XX(hex), where XX is as follows:

LSB = Bit 0 = Document at throat
= Bit 1 = Document Jam **(DOES NOT WORK, ALWAYS 0)**
= Bit 2 = Power on self test failure **(CAN NOT CHECK THIS)**
= Bit 3 = Not used
= Bit 4 = Transport cleared
= Bit 5 = Transport enabled
= Bit 6 = Q.C Test Mode
MSB = Bit 7 = Parity bit **(ALWAYS 0)**

When a Bit = 1, it is set (active)
When a Bit = 0, it is not active

22. CHECK READER'S VERSION (V):

Ctrl R ESC V Ctrl R E

NOTE: The reader will return a version string